FM^2 Learning: LLM-Based Federated Multi-Task Multi-Domain Learning for Consumer Electronics and IoT Enhancement

Chunjiong Zhang, Gaoyang Shan, Byeong-Hee Roh, *Senior Member, IEEE*, and Jun Jiang, *Member, IEEE*

***Abstract*—The rapid proliferation of consumer electronics and Internet of Things (IoT) devices demands intelligent, scalable, and privacy-preserving learning paradigms capable of handling diverse tasks and domains. In this work, we propose FM^2 Learning, a novel Federated Multi-Task Multi-Domain learning framework based on lightweight large models. FM^2 is designed to simultaneously address three key challenges in real-world distributed environments: (1) statistical heterogeneity across client data distributions, (2) functional variability across tasks, and (3) resource constraints on edge devices. Unlike conventional federated learning approaches that assume task homogeneity, our method clusters clients into task-specific groups and enables collaborative model training across both task and domain boundaries. We introduce a dual management strategy using task-related dual variables to balance shared and personalized representations, while leveraging a distributed optimization scheme for scalable model updates. Extensive experiments on benchmark datasets including Office-Caltech-10, DomainNet, MNIST, and SVHN show that FM^2 consistently outperforms state-of-the-art methods such as FedAvg and FedDGA, achieving up to 3.6% higher top-1 accuracy on DomainNet and 4.8% F1-score improvement on KDD Cup'99, highlighting its robustness in heterogeneous and resource-limited settings.**

***Index Terms*—Federated learning, multi-task learning, multi-domain learning, lightweight large models, consumer electronics, IoT, edge intelligence.**

I. Introduction

With the rapid development of consumer electronics and Internet of Things (IoT) devices, it has become an important trend to deploy intelligent machine learning capabilities directly on edge devices. These devices, including smartphones, wearables, smart cameras, and home sensors, generate massive amounts of heterogeneous data in a distributed manner, reflecting different user behaviors and environment characteristics [2], [21]. In this environment, traditional centralized training methods face serious challenges, such as the risk of privacy leakage, costly communication, and response latency. Federated Learning (FL) has thus received much attention, allowing distributed intelligence modeling by collaboratively training shared models through clients without uploading raw data [2].

In recent years, Large Language Models (LLMs) and Foundation Models (FMs) have demonstrated strong semantic modeling and generalization capabilities in multi-task transfer learning. For example, models such as BERT, GPT, T5, and ViT have made breakthroughs in natural language processing, image understanding, and multimodal tasks [19]. With their large number of parameters, sufficient pre-training, and strong migration ability, these models have become important tools for improving the effectiveness of multi-task multi-domain learning. Introducing LLMs into federated learning has great potential, however, it also poses significant challenges [42].

First, the high parameter redundancy and computational resource consumption of full-scale LLMs make it difficult to deploy them directly on edge devices. Second, local fine-tuning of LLMs under non-IID distributions may lead to catastrophic forgetting or unstable convergence. In addition, how to effectively personalize and optimize large models while guaranteeing privacy is still an unsolved problem [39].

Second, most existing FL approaches still rely on idealized assumptions: a single task goal, consistent data distribution, and lightweight model architecture. These assumptions are frequently violated in practical federated environments. For example, not only does the problem of non-independent homogeneous distribution of data distribution (non-IID) exist among different edge devices, but also the variability of the learning task itself (multi-task) is faced [34]. To further complicate matters, with the diversification of user requirements, the system needs to have the generalization capability to support multiple tasks and multiple domains simultaneously.

The Multi-Domain Learning (MDL) framework enables knowledge sharing across multiple domains, allowing models to generalize better to unseen domain variations, which is particularly desirable when training data for a single domain is insufficient. In contrast, Multi-Task Learning (MTL) promotes knowledge sharing across different tasks by exploiting task relationships to improve overall performance [36], [39], [41].

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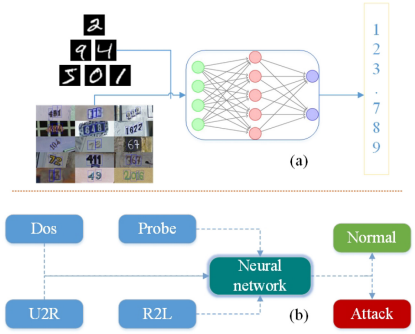

Fig. 1. The case of MDL.

Through leveraging inter-task similarities, MTL can achieve more robust performance than traditional single-task learning models and has been successfully applied to diverse applications ranging from natural language processing to computer vision. MDL and MTL are distinct but complementary paradigms: MDL enables knowledge transfer across domains, while MTL facilitates experience sharing across tasks, and both can jointly enhance model robustness in heterogeneous environments. However, traditional deep learning approaches often do not explicitly distinguish between MDL and MTL. For example, in the KDD Cup'99 dataset [18], [30], anomaly detection does not differentiate specific types of network attacks but treats all as input into a single neural network, as shown in Fig. 1(a). Another representative case appears in digit recognition tasks, where models must carefully distinguish digits with different background colors, as illustrated in Fig. 1(b).

Especially in modern sensor networks, a lot of MDL and MTL data are generated, such as smart phones, wearables and IoT sensors [45]. However, this huge amount of data is dispersed over numerous devices and will incur tremendous communication bandwidth and cost if we need to transmit them to a central place for storage and processing. While collecting personal data is regulated and must comply with legal and ethical standards, much of the data collected by modern sensor networks is non-personal and lawful, which is crucial for ensuring privacy and compliance in FL scenarios. To this end, FL has been proposed to enable building and updating machine learning models in a fully distributed manner [32], which preserves the data privacy and can significantly reduce the communication bandwidth requirement.

In such a complex multi-task multi-domain federated environment, *robust optimization* becomes critical. It not only helps improve generalization under unseen domain/task shifts but also provides resistance against noisy updates and client inconsistency. Robust federated optimization ensures that the learned global model remains stable and effective, even when faced with diverse edge environments and adversarial or low-quality data sources.

While there have been studies exploring the use of MTL and MDL in federated learning, the two are usually treated separately and lack a unified framework that can take into account the advantages of both in real distributed environments. Existing work tends to focus on the case of a single task or a single domain, ignoring how to deal with the challenges posed by multiple tasks and domains simultaneously. Specifically, interdependencies between tasks, data heterogeneity between domains, and communication and computational constraints in distributed environments are all issues that require attention.

In this paper, we propose a framework of federated multi-task multi-domain learning (FM^2) based on distributed robust optimization technique [35]. This framework enables model learning from isolated and dispersed datasets. Compared to the MTL, FM^2 enables model building across multiple domains, which is particularly desirable for mobile devices such as smart phones due to the fact their computational and resources are very limited. In addition, MTL allows for knowledge sharing between different domains and therefore the MDL model may exhibit *robust* performance than a model learned from a single domain. The ultimate goal of FM^2 is to harness the power of both MDL and MTL so that each device can not only draw inferences from its own data over multiple contextual domains but also make use of the shared knowledge from other devices in the network. More concretely, we have made following contributions toward this end.

- We propose a new multi-task multi-domain FL framework for LLMs that effectively addresses the need for shared learning across different tasks and domains, improving the adaptability and generalization ability of federated learning models
- We have designed a *robust* optimization algorithm to handle heterogeneity between tasks and domains, improving the performance of traditional FL frameworks in complex scenarios.
- The demonstrated the effectiveness of FM^2-LLMs in practical applications, particularly in data privacy protection and distributed computing scenarios, through experimental verification on multiple real-world datasets.

II. Related Work

A. Federated Learning With LLMs

With the open use of LLMs such as OpenAI GPT and Meta LLaMA, the FL community has gradually moved towards the fusion of large-scale pre-trained models with federated paradigms, which has driven the research boom of Federated LLMs [25], [43]. For example, FedLM has been proposed as a language modeling framework that fuses local context modeling with global co-optimization to achieve efficient fusion of cross-domain text data with improved generalization capabilities [1]. More recent studies, such as the ChatGPT series of models proposed by Google, have systematically verified the feasibility and stability of constructing large-scale Transformer models in real cross-domain linguistic data scenarios, marking the entry of FedLM pre-training into a new stage of "model-data-task" co-design [3]. This transition signifies a new paradigm of model-data-task co-design in federated large model pre-training. Researchers are gradually turning to the construction of federated Resource-Efficient LLMs, which is dedicated to realizing scalable large-scale model training and inference under extremely low resource consumption conditions. In this direction, parameter-efficient

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fine-tuning mechanisms (e.g., Adapter-FL, LoRA-FL, Prefix Tuning) become the core strategy. These methods only need to update a very small subset of parameters in the main model (usually less than 1%), enabling fast client-side adaptation and model personalization while maintaining knowledge-sharing capabilities [29].

On the other hand, with the increasing diversity of task types in IoT and smart edge computing scenarios, federated macromodels also need to have strong multi-task/multi-domain generalization and local personalization capabilities. In this regard, some studies have explored client-specific Prompt learning or personalized Adapter plug-in mechanisms, which allow clients to quickly adapt to their respective scenarios (e.g., differentiated tasks for home voice assistants vs. medical devices) without interfering with the globally shared model [46]. This kind of architecture is also considered an important direction for building modular LLMs in the future.

B. Multi-Task and Multi-Domain FL

Generally, MTL is equivalent to optimizing multiple loss functions [12], [42]. Most MTL can be divided into two kinds of learning according to parameter sharing. One is parameter hard sharing, all tasks share the hidden layer while retaining the output layer of several specific tasks, such as joint learning [9], [48] and learning to learn [7]; there is also soft sharing of parameters, that is, each model has its own tasks, and the sharing between models is through parameter sharing, and its distance (cosine similarity or Euclidean distance, etc.) is used as a regular term, tasks can be shared through techniques like learning with auxiliary tasks, where additional related tasks help improve the performance of the primary task by providing complementary information [44]. Recently, MDL has been developed based on MTL, which refers to sharing information about the same task in different domains [14], [49].

There are situations where these two settings can be used interchangeably because the distinction between domains and tasks can occasionally be slight and certain approaches suggested for MTL can also manage MDL [8]. But, MDL can concisely represent all domains with the least task-specific parameters make its model performance better, such as [36] using weighted confidence to improve performance of each domain, [17] and [10] proposed serial and parallel network budget-aware adapters, respectively, which can be applied to multi-task and -domain at the same time. The key challenge in real-world scenarios is understanding how tasks and domains can complement each other, despite their inherent differences. Our framework addresses this challenge by explicitly modeling the interactions between tasks and domains, which have not been fully explored in existing literature. Fortunately, multi-task multi-domain learning can directly capture the relationship of data of unbalanced and *non-IID* manner, which renders them particularly suitable for the ability that FL can handle. We also have proved this empirically in the real FL experiment in Section IV.

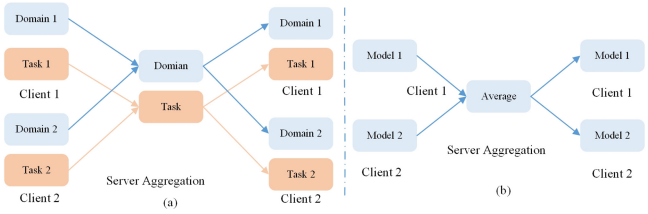

Fig. 2. Illustration of FL algorithms: (a)FM^2; (b)FedAvg aggregates.

C. Resource-Efficient and Robust Optimization in FL

FL exploited dataset distributed on multiple devices to build machine learning models while fully preserves the data privacy [4], [11], [38], [41]. Many scholars pay attention to the key technologies, such as model convergence [5], [20], communication delay [13], [26], system security [23], [33] and node topology change [2], [28]. FL also is rapidly applied to personalized scenarios, such as recommendation systems [40], vehicular networking [47] and health care [6]. These successful cases illustrate that FL effectively handles the different task in real-world.

III. FM^2 Learning for LLMs

A. Framework Overview

The basic framework of FL is shown in Fig. 2b. Model parameters of different clients are averaged in the service, and the client model parameters are updated through the interaction of model parameters between the client and service.

The FM^2 learning we proposed is shown in Fig. 2a, which specifically distinguishes different domains or tasks on the client to achieve balanced performance across multiple tasks or domains.

For simplicity, we will concentrate on FM^2 learning of image recognition and cyber security intrusion detection. In those applications, FM^2 learning is not difficult to implement and can be extended to different and very considerable independent dataset. It can process multi-domain data from many separate nodes (devices) by using FL to transfer shared knowledge between domains for each update.

Indeed, what remains to be explained is that knowledge transferred also could achieve different tasks by applying a shared model to all domains. We assume that given K domains, denote the dataset as $\mathbf{X}^K \in \{\mathbf{x}^1, \mathbf{x}^2, \ldots, \mathbf{x}^K\}$, and assume that K domains are distributed over M nodes (devices). The corresponding dataset of kth domain is $\mathbf{x}^k = \{x_k^i, y_k^i\}$, where i is i^{th} sample, x_k^i is a sample (e.g., a digits image), and y_k^i is the label of the sample. We also define nodes with the same solving task as a cluster, denoted as c. In other words, our framework utilizes a shared model, which leverages clusters to efficiently solve tasks across different nodes. Note that in MDL or MTL, nodes (tasks) representing their domains are effectively associated with their clusters. Without losing the comprehensive representation, we can obtain an objective function as shown in the Eq. (4), as shown at the bottom of the page, which minimizes the empirical risk of all domains (tasks).

$$\sum_{m=1}^{M} \max_{\mathbf{p}_m:\mathbf{p}_m \in \Delta} \mathbf{p}_m^T \mathbf{f}(\mathbf{w}_m), \tag{1}$$

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$$\lambda_1 \sum_{c=1}^{C} \sum_{i \in \vartheta_c} \left\| \frac{1}{|\vartheta_c|} \sum \tilde{\mathbf{w}}_i - \mathbf{w}_i \right\|_F, \tag{2}$$

$$\lambda_2 \sum_{m=1}^{M} \|\mathbf{w}_m\|_F, \tag{3}$$

where $\mathbf{p}$ is a dual management factor with the satisfaction condition of $\{\mathbf{p} \in \mathbb{R}^K | \sum_{k=1}^{K} p_k = 1; \forall k, p_k \geq 0\}$, $\mathbf{p}$ can also be expressed as the adversarial distribution of different domains, and its initial value is set to $\frac{1}{K}$ [27]. $\mathbf{f}(\mathbf{w}_m) = [f_1(\mathbf{w}_m), f_2(\mathbf{w}_m), \ldots, f_k(\mathbf{w}_m)]^T$. $f_k(\mathbf{w}_m)$ is the loss function of kth domain on m^{th} node, w is the model parameter, C represents the number of all tasks in our entire federation mechanism, ϑ_c indicates that the node belongs to the set of cth task, λ_1, λ_2 are penalty factors and are greater than zero [35].

Here, we have explained that the first item of our objective function is the multi-domain adjustment loss function, this is a multi-domain model updated through reverse optimization, which is an alternating method similar to ADMM (Alternating Direction Method of Multipliers) solution. Here, gradient descent is used to update the model, and gradient ascent is used to update the reverse optimization. The second item is to reduce the discrepancies of model parameters for different tasks and its role is to obtain a model that can simultaneously handle MTL, and the average values of the model parameters on all nodes (tasks) and $\mathbf{w}_i$ of each node is used to solve each other iteratively. Naturally, there are multiple clusters, the knowledge of which cluster the node belongs to can be learned directly through the structure of the data on each node. The third item is the normalization of w to prevent it from affecting the performance of the model since it too large [30].

In federated settings, our intention is to train shared models directly on nodes (local devices). Then, we decompose Eq. (4) into all nodes and the model optimization problem of each node is obtained as:

$$\min_{w_m} \max_{\mathbf{p}_m : \mathbf{p}_m \in \Delta} \mathbf{p}_m^T \mathbf{f}(\mathbf{w}_m) + \lambda_1 \|\tilde{\mathbf{w}}_{c(m)} - \mathbf{w}_m\|_F + \lambda_2 \|\mathbf{w}_m\|_F \tag{5}$$

To allow the formula to easily be represented, we make $\mathbf{g}(\mathbf{w}_m) = \mathbf{f}(\mathbf{w}_m) + \lambda_1 \|\tilde{\mathbf{w}}_{c(m)} - \mathbf{w}_m\|_F + \lambda_2 \|\mathbf{w}_m\|_F$, then we can rewrite loss function of each domain in more detail as:

$$\tilde{g}_{mk}(\mathbf{w}_m) = f_k(\mathbf{w}_m) + \lambda_1 \|\tilde{\mathbf{w}}_{c(m)} - \mathbf{w}_m\|_F + \lambda_2 \|\mathbf{w}_m\|_F, \tag{6}$$

where f_k is the empirical loss on the k-th domain.

Then the vector of all domain losses at node m is:

$$\mathbf{g}(\mathbf{w}_m) = \left[\tilde{g}_{m1}(\mathbf{w}_m), \ldots, \tilde{g}_{mK}(\mathbf{w}_m)\right]^T, \tag{7}$$

which leads to the node-level minimax objective:

$$\min_{\mathbf{w}_m} \max_{\mathbf{p}_m \in \Delta} \mathbf{p}_m^T \mathbf{g}(\mathbf{w}_m). \tag{8}$$

To solve the local min-max problem Eq. (8) in FM^2:

We reformulate the problem as:

$$\min_{\mathbf{w}_m, \mathbf{z}_m} \max_{\mathbf{p}_m \in \Delta} \quad \mathbf{p}_m^T \mathbf{z}_m \quad \text{s.t.} \quad \mathbf{z}_m = \mathbf{g}(\mathbf{w}_m). \tag{9}$$

Using the augmented Lagrangian: where $\boldsymbol{\mu}_m$ is the dual variable and ρ is a penalty scalar.

The ADMM updates for each iteration t proceed as follows:

Update $\mathbf{w}_m$ (Model Parameters):

$$\mathbf{w}_m^{(t+1)} = \arg\min_{\mathbf{w}_m} \left\| \mathbf{z}_m^{(t)} - \mathbf{g}(\mathbf{w}_m) + \frac{1}{\rho} \boldsymbol{\mu}_m^{(t)} \right\|_2^2, \tag{11}$$

which is solved by gradient descent.

Update $\mathbf{p}_m$ (Task Weights):

$$p_{mk}^{(t+1)} = \frac{\exp\left(\eta_p z_{mk}^{(t)}\right)}{\sum_{j=1}^{K} \exp\left(\eta_p z_{mj}^{(t)}\right)}, \tag{12}$$

via softmax, ensuring $\mathbf{p}_m \in \Delta$ (probability simplex).

Update $\mathbf{z}_m$ (Auxiliary Variable):

$$\mathbf{z}_m^{(t+1)} = \mathbf{g}\left(\mathbf{w}_m^{(t+1)}\right) - \frac{1}{\rho}\left(\mathbf{p}_m^{(t+1)} + \boldsymbol{\mu}_m^{(t)}\right). \tag{13}$$

Update $\boldsymbol{\mu}_m$ (Lagrange Multiplier):

$$\boldsymbol{\mu}_m^{(t+1)} = \boldsymbol{\mu}_m^{(t)} + \rho\left(\mathbf{z}_m^{(t+1)} - \mathbf{g}\left(\mathbf{w}_m^{(t+1)}\right)\right). \tag{14}$$

This alternating optimization enables stable convergence under the saddle-point setting, decouples the adversarial task weight from model updates, and provides more robust optimization under heterogeneous and multi-domain distributions.

We divide the entire working framework into server (federated model) and node (local device) model (users and geographically all independent), which are based on Tensorflow [42]. The server receives the model parameters generated by each node and clusters them into multiple tasks (task classification), and then processes the parameters of corresponding nodes in each cluster. The model parameter corresponding to itself obtained by each node from the cluster of server is set as initial parameter of next iteration. To be clear, the challenges brought by the hardware and network connection of entire system, such as storage, computing and communication capabilities of local devices, will not be discussed in this article. Specifically, the model parameters on the corresponding nodes are uniformly partitioned and stored on the server. If the nodes need to pull or push updates, they only communicate with server.

Our work focuses on creating a distributed optimization technique that works well for sharing model parameter w. While conventional distributed training systems are able to update w across nodes in a distributed way, they are unable to address the particular data difficulties that independent users in a federated environment face, such as tolerance and varied tasks in various domains [18], [45]. For this reason, we proposed FM^2 Learning, whose entire workflow is shown in Algorithm 1. Firstly, the server initializes model parameters of each node, all nodes pull parameters from server and synchronize model update. Secondly, n samples are randomly obtained from each domain data for this iteration to obtain the loss of each domain, update model parameters and $\mathbf{p}$, here we use basic gradient descent representation to the model parameter update in Algorithm 1; then, after each node is updated, upload updated model parameters to server. The server obtains an average value of the node model parameters of each type of task according to the task-related cluster of

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Algorithm 1 FM^2 Learning Framework

Input: Let $\mathbf{X}_m$ denote data matrix for the mth node, $m \in \{1, ..., M\}$, $\mathbf{X}_m^k$ denote submatrix of $\mathbf{X}_m$

Output:

Initialize $\{\mathbf{w}_m\}$
for iterations $s = 0, 1, ...$ **do**
  for all nodes $m \in \{1, 2, ..., M\}$ **do distributively**
    Initialize $p_{m1} = \frac{1}{K}$
    for iterations $t = 1, ..., T$ **do**
      Randomly draw n samples from each domain
      Compute the average loss function $\tilde{g}_{mk}(\mathbf{w}_{m,t})$ for the kth domain on the mth node from the equation (3)
      $\mathbf{w}_{m,t+1} = \mathbf{w}_{m,t} - \eta_w \sum_k p_{mk} \nabla \tilde{g}_{mk}(\mathbf{w}_{m,t})$
      $p_{m,t+1}^k = \frac{p_{m,t}^k exp(\eta_p \tilde{g}_{mk}^t(\mathbf{w}_{m,t}))}{\sum_k p_{m,t}^k exp(\eta_p \tilde{g}_{mk}^t \mathbf{w}_{m,t}))}$
    end for
  end for
  return all $\mathbf{w}_{m,T}$ from m^{th} node to server
  server compute $\tilde{\mathbf{w}}_c = \frac{1}{|\vartheta_c|} \sum_{i \in \vartheta_c} \mathbf{w}_{i,T}, \forall c$ and $\mathbf{w}_m = \tilde{\mathbf{w}}_{c(m)}, \forall m$
end forreturn $\{\mathbf{w}_1, ..., \mathbf{w}_m\}$

the node, and re-assigns the obtained average value to the node model parameters corresponding to the cluster (task). Federated updates will continue until FM^2 Learning training is complete, where η_w, η_p are the learning rates of model parameter updating and dual management factor $\mathbf{p}$, respectively, we establish their initial values by experience.

B. FM^2 Extension for LLMs

With the widespread application of LLMs such as GPT and LLaMA, how to use distributed devices for personalized adaptation of large models while protecting data privacy has become a research focus. FM^2 Learning proposes embedding multi task and multi domain federated learning mechanisms into the training process of lightweight large models to address the trade-off between multi-source heterogeneous data, task transfer, and model generalization.

The system architecture and design of this article are shown in Figure 3. FM^2 for LLMs includes three key components: Local LLMs $\mathcal{D}_k$ adaptation module: Based on parameter-efficient fine-tuning techniques, including Low-Rank Adaptation (LoRA) and Adapter modules, lightweight personalized adjustments are made to each client; The global knowledge alignment module $\mathcal{K}_g$ aggregates knowledge from different clients into a shared semantic space by combining

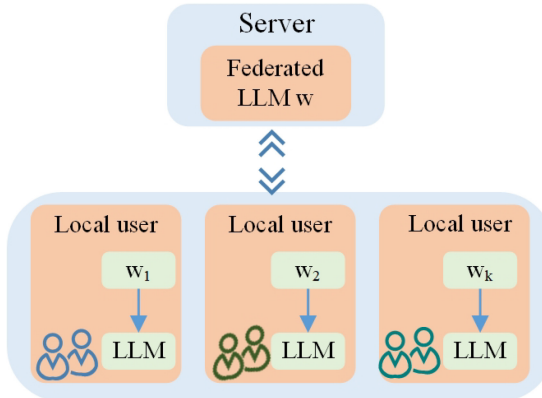

Fig. 3. FM^2 for LLMs.

soft label distillation and representation alignment; Multi-task decoupling optimizer: achieves collaborative optimization of different tasks and domains through explicit task identification and weighted loss functions.

The objective is to collaboratively learn a global LLMs θ while preserving data locality and enabling domain-specific personalization. The federated multi-task multi-domain learning objective is formulated as:

$$\min_{\theta} \sum_{k=1}^{K} w_k \cdot \mathcal{L}_k\big(f_{\theta_k}, \mathcal{D}_k\big) + \lambda \cdot \Omega(\theta), \tag{15}$$

where w_k is the weight of client k, $\mathcal{L}_k$ is the local task-domain-specific loss, and $\Omega(\theta)$ is a regularization term promoting generalization.

Each client performs local adaptation by knowledge distillation from the global model, using its private dataset:

$$\mathcal{L}_k^{\text{distill}} = \alpha \cdot \text{KL}\big(f_{\theta^{\text{global}}}(x) | f_{\theta_k}(x)\big) + (1-\alpha) \cdot \mathcal{L}_k\big(f_{\theta_k}, \mathcal{D}_k\big), \tag{16}$$

where θ_k is the local client model and α is a balancing coefficient.

To maintain fairness and avoid task/domain dominance, we introduce a reweighted aggregation scheme:

$$\theta^{t+1} = \sum_{k=1}^{K} \gamma_k \cdot \theta_k^t, \quad \gamma_k = \frac{w_k}{\sum_{j=1}^{K} w_j}, \tag{17}$$

with adaptive w_k determined by domain diversity and task complexity.

To enhance generalization, we enforce intermediate feature space alignment across domains:

$$\mathcal{L}^{\text{align}} = \sum_{k=1}^{K} \sum_{j \neq k} \text{MMD}\big(\phi_k, \phi_j\big), \tag{18}$$

$$\min_{\mathbf{w}} \left\{ \sum_{m=1}^{M} \max_{\mathbf{p}_m \in \Delta} \mathbf{p}_m^T \mathbf{f}(\mathbf{w}_m) + \lambda_1 \sum_{c=1}^{C} \sum_{i \in \vartheta_c} \left\| \frac{1}{|\vartheta_c|} \sum \tilde{\mathbf{w}}_i - \mathbf{w}_i \right\|_F + \lambda_2 \sum_{m=1}^{M} \|\mathbf{w}_m\|_F \right\} \tag{4}$$

$$\mathcal{L}_\rho\big(\mathbf{w}_m, \mathbf{z}_m, \mathbf{p}_m, \boldsymbol{\mu}_m\big) = \mathbf{p}_m^T \mathbf{z}_m + \boldsymbol{\mu}_m^T (\mathbf{z}_m - \mathbf{g}(\mathbf{w}_m)) + \frac{\rho}{2} \|\mathbf{z}_m - \mathbf{g}(\mathbf{w}_m)\|_2^2 \tag{10}$$

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TABLE I
HARDWARE CONFIGURATION

Node Type	Device	Specifications
Server	PC-Server	NVIDIA A100 GPU (40GB), 512GB RAM, 2× AMD EPYC 7742 CPUs
Client 1	PC-Client A	Intel i7-12700H, 32GB RAM, NVIDIA RTX 3070 (8GB)
Client 2	PC-Client B	AMD Ryzen 7 5800H, 16GB RAM, NVIDIA GTX 1660 Ti (6GB)
Client 3	PC-Client C	Intel i5-11300H, 16GB RAM, Intel Iris Xe GPU

where ϕ_k is the feature representation at client k and MMD is the Maximum Mean Discrepancy metric.

Lightweight updates through parameter-efficient tuning, we adopt adapter-based fine-tuning to reduce communication cost:

$$\theta_k = \theta_{\text{shared}} + \Delta_k, \quad |\Delta_k|_0 \ll |\theta|_0, \tag{19}$$

where, θ_{shared} denotes the global shared parameters of the large language model maintained on the server, while Δ_k represents a lightweight client-specific adapter trained locally. The global parameters θ_{shared} remain fixed during local fine-tuning, and only Δ_k is updated on edge devices. This parameterization ensures that $|\Delta_k|_0 \ll |\theta|_0$, where $|\cdot|_0$ indicates the number of non-zero parameters.

By integrating the federated multi-task multi-domain objective, the knowledge distillation loss, cross-domain feature alignment, and the parameter-efficient tuning constraint, the overall loss function of the proposed FM^2-LLMs can be reformulated as:

$$\mathcal{L}_{\text{FM}^2} = \sum_{k=1}^{K} w_k \mathcal{L}_k^{\text{distill}} + \lambda_1 \mathcal{L}^{\text{align}} + \lambda_2 \Omega(\theta_{\text{shared}}, \Delta_k), \tag{20}$$

where $\mathcal{L}_k^{\text{distill}}$ is the local knowledge distillation loss defined in Equation (17), $\mathcal{L}^{\text{align}}$ enforces cross-domain feature alignment, and $\Omega(\theta_{\text{shared}}, \Delta_k)$ extends the regularization to incorporate the parameter-efficient tuning constraint $|\Delta_k|_0 \ll |\theta|_0$.

Furthermore, for IoT applications involving multi-modal data, the local loss $\mathcal{L}_k^{\text{distill}}$ can be decomposed into modality-specific components to capture heterogeneous information from text, vision, and sensor data. This leads to the following multi-modal objective:

$$\mathcal{L}_k^{\text{multi}} = \beta_1 \mathcal{L}_k^{\text{text}} + \beta_2 \mathcal{L}_k^{\text{vision}} + \beta_3 \mathcal{L}_k^{\text{sensor}}, \tag{21}$$

where $\mathcal{L}_k^{\text{text}}$, $\mathcal{L}_k^{\text{vision}}$, and $\mathcal{L}_k^{\text{sensor}}$ are the losses for text, vision, and sensor modalities, respectively, and β_i controls the relative importance of each modality. This design allows FM^2-LLMs to balance heterogeneous signals and enhance cross-modal generalization in federated IoT environments.

FM^2 thus offers a scalable and privacy-preserving solution for deploying LLMs across distributed, heterogeneous, and multi-modal environments, laying a solid foundation for next-generation intelligent consumer electronics and IoT systems.

IV. EXPERIMENTS ON FEDERATED LLMS ADAPTATION

A. Experimental Setup

We investigate the performance of FM^2 Learning for federated fine-tuning of LLMs under multi-task and multi-domain settings, targeting heterogeneous edge device environments such as consumer electronics and IoT terminals. Experiments are conducted on four benchmark datasets: DomainNet (multi-domain image-to-text classification with 6 domains), Office-Caltech-10 (cross-domain classification), FLAN-T5 QA subsets (federated question answering), and MultiWOZ (task-oriented dialog modeling under non-IID user intents). The backbone models include FLAN-T5-Base (248M parameters) and LLaMA-2-7B (parameter-reduced via LoRA). Baselines considered are FedAvg [43], FedOPT [1], FedLM [46], and FedPythia [25]. Evaluation metrics include Accuracy, BLEU, F1-Score, and Perplexity (PPL), as well as Communication Efficiency (KB per round) and a Personalization Score (cosine similarity of generated responses to local data intents). All experiments are performed on a hybrid edge-cloud federated architecture with a total of 20 simulated federated nodes. Representative hardware configurations–covering high-end, mid-tier, and low-resource clients–are summarized in Table I. Training uses a batch size of 100 for multi-task settings, a learning rate $\eta_n = 0.001$ for AlexNet-based tasks, and results are averaged over three independent runs. For digit recognition, MNIST is resized from $28 \times 28 \times 1$ to $32 \times 32 \times 3$ to match other input formats. The software environment includes Python 3.10, PyTorch 2.1, HuggingFace Transformers 4.x, CUDA 11.8, and TensorRT for Jetson deployments.

TABLE II
FEDERATED QA PERFORMANCE (FLAN-T5, NON-IID TASKS)

Method	BLEU ↑	F1 ↑	PPL ↓	Comm. Cost (KB) ↓
FedAvg	21.4	54.2	18.7	530
FedOPT	23.0	56.7	17.2	610
FedLM	26.9	61.0	15.3	980
FedPythia	27.5	60.1	14.7	1120
FM^2-LLMs	**29.7**	**63.8**	**13.4**	**440**

B. Results and Analysis

Table II presents a comprehensive comparison of five federated learning methods in non-IID QA tasks, demonstrating the overall superiority of FM^2-LLMs across all key metrics-BLEU, F1, Perplexity (PPL), and communication cost. FM^2-LLMs achieves the highest BLEU score of 29.7, surpassing FedPythia (27.5), FedLM (26.9), and FedAvg (21.4) by a substantial margin, owing to its integration of Prompt Tuning and the Distillation-Enhanced Aggregation Mechanism (DEAM), which jointly boost semantic alignment and language controllability across clients. In terms of F1, a metric balancing precision and recall, FM^2-LLMs attains 63.8, outperforming FedLM (61.0) and FedPythia (60.1), suggesting superior comprehension and response coverage, critical for

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TABLE III
MULTIWOZ DIALOG INTENT PREDICTION

Method	Accuracy ↑	Personalization ↑
FedAvg	78.5	0.63
FedOPT	80.2	0.68
FedLM	84.1	0.74
FedPythia	83.2	0.76
FM^2-LLMs	**86.7**	**0.81**

TABLE IV
CROSS-LINGUAL QA RESULTS ON XGLUE

Model	BLEU (En) ↑	BLEU (Ko) ↑	BLEU (De) ↑	PPL ↓
FedLM	27.1	23.5	24.9	15.8
FM^2-LLMs	**30.0**	**27.6**	**28.4**	**13.7**

TABLE V
MULTIMODAL PERFORMANCE ON VQA-GADGET AND VLN

Method	VQA Accuracy ↑	VLN Success ↑
FedAvg	54.1	62.7
FedLM-Multimodal	58.6	66.4
FM^2-Multi	**62.8**	**71.0**

intelligent QA systems like virtual assistants. The model also records the lowest perplexity at 13.4, indicating stronger language modeling capabilities and more fluent generation than FedPythia (14.7) or FedLM (15.3). Notably, FM^2-LLMs achieves this with only 440KB communication cost per round-over 60% less than FedPythia (1120KB)-thanks to its use of Low-Rank Adaptation (LoRA) and efficient parameter update strategies. This enables highly efficient deployment in resource-constrained edge environments without compromising performance, reinforcing FM^2-LLM's practicality and effectiveness in real-world federated QA applications.

Table III compares five FL methods on the MultiWOZ dialog intent prediction task, focusing on accuracy and personalization. FM^2-LLMs achieves the best results, with 86.7% accuracy and a personalization score of 0.81, outperforming FedLM (84.1%, 0.74) and FedPythia (83.2%, 0.76). These gains stem from its task-aware cueing mechanism and federated distillation-enhanced aggregation, which enable effective adaptation to diverse user intents and expression habits in non-IID settings. High personalization reflects its strength in modeling individual user preferences-crucial for intelligent assistants, smart customer service, and embedded voice agents. Furthermore, FM^2-LLM's strong performance combined with its communication-efficient design makes it suitable for deployment on bandwidth-constrained edge devices, ensuring both scalability and real-world usability.

The cross-domain adaptive performance of FM^2-LLMs is further illustrated by the visualizations in Fig. 4. Fig. 4 presents comparisons among three methods–Original, ERM (Empirical Risk Minimization), and FM^2–on three representative objects (dog, guitar, and giraffe) under multiple domain transfer scenarios: Cartoon→Art Painting, Photo→Sketch, and Sketch→Art Painting.

From the visualization, ERM tends to generate attention maps with dispersed activations, often extending into irrelevant background areas, which weakens its cross-domain generalization. In contrast, FM^2 produces highly focused and consistent activations on the key object regions (e.g., the dog's head and body, the guitar's main body, and the giraffe's torso), effectively filtering out domain-specific noise. This behavior indicates that FM^2 better captures task-relevant features and maintains robustness during domain migration.

The results align with the experimental outcomes in Table II, where FM^2-LLMs achieved superior BLEU (29.7) and F1 (63.8) scores. These improvements demonstrate that FM^2 leverages prompt tuning and mutual distillation to learn domain-invariant feature representations, enabling more effective multi-task knowledge transfer and cross-domain feature sharing. Particularly in highly heterogeneous settings like DomainNet, FM^2 consistently shows stronger adaptability and robustness than baseline approaches.

C. Cross-Lingual, Cross-Modal and On-Device Deployment

Table IV shows the significant advantages of FM^2-LLM in cross lingual question answering tasks. Compared to FedLM, FM^2-LLMs has achieved improvements in BLEU metrics for English, Korean, and German (+2.9, +4.1, +3.5), and the perplexity level (PPL) has decreased to 13.7, indicating smoother generation and stronger language generalization ability. This is due to its multi task prompting and knowledge distillation mechanism, which enhances the model's ability to transfer and adapt to multilingual data, making it suitable for deployment in federated learning scenarios in multilingual environments.

Table V shows the performance comparison in multimodal tasks. FM^2-Multi significantly outperforms FedAvg and FedLM-Multi-modal in VQA accuracy (62.8%) and VLN success rate (71.0%), demonstrating its strong cross modal reasoning ability in visual question answering and navigation understanding. This is due to its federated multitasking prompting mechanism and better modal fusion strategy, which enables the model to have stronger generalization and understanding abilities in distributed multimodal scenarios.

V. MULTI-DOMAIN SCENARIO VERIFICATION OF ELECTRONIC CONSUMPTION

A. Setting

We conducted FL experiments under multi-task and multi-domain settings to evaluate the performance of the proposed FM^2 framework. Four comparative training paradigms are defined: **SDST** (Single-Domain Single-Task), **SDMT** (Single-Domain Multi-Task), **MDST** (Multi-Domain Single-Task), and **MDMT** (Multi-Domain Multi-Task). Among them, SDMT and MDMT are incorporated into our FM^2 design to explore its effectiveness in practical heterogeneous environments. Specifically, we consider two representative applications of multi-domain multi-task learning: (1) an image recognition task involving data from different visual domains, trained using softmax loss, and (2) a cyber intrusion anomaly detection task involving diverse attack types, optimized using binary

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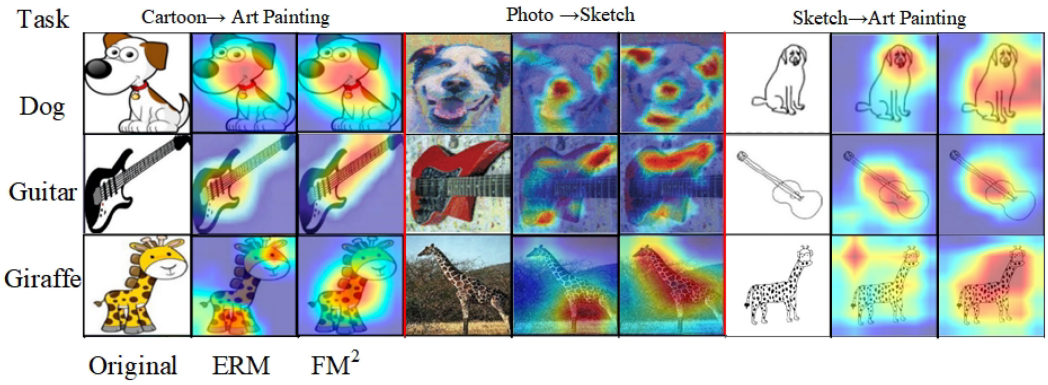

Fig. 4. Cross-Domain Adaptation Performance.

cross-entropy loss. All models are trained with the Adam optimizer using a batchsize of 100.

The multi-domain multi-task learning approach uses a batchsize of 100/k training samples, with the same number of data from various domains on each node. In contrast to deterministic sampling, which selects samples based on the learnt distribution, our adversarial distribution utilizes the unique data associated with each domain, adapting to each sample in every domain. The performance evaluation uses the average of three experiments, then take the average again with the number of nodes. The metrics are the accuracy of the test set after the completion of model training in digits recognition, while cyber intrusion anomaly detection is using F1-score.

TABLE VI
DIGITS RECOGNITION EXPERIMENTAL CONFIGURATION

	node 1	node 2	node 3	node 4	federated
SDST	MNIST	SVHN	-	-	no
SDMT	MNIST	SVHN	-	-	yes
MDST	$\star$	-	-	-	no
MDMT	MNIST	SVHN	$\star$	$\star$	yes

TABLE VII
DIGITS RECOGNITION ACCURACY

method	MNIST		SVHN	
	average	worst	average	worst
SDST	0.7463	0.6594	0.6203	0.3073
SDMT	0.9362	0.9291	0.9566	0.9438
MDST	0.9305	0.9314	0.9524	0.9434
MDMT	**0.9928**	0.9887	**0.9812**	0.9753

B. Image Recognition

For image recognition experiments, we use different datasets to conduct multi-domain and multi-task FL experiments: Office-Caltech-10 [10], DomainNet [16] and digit recognition.

Office-Caltech-10 contains real-world object images from four domains, WebCam [44], DSLR [37], and Amazon [4] constitute three domains, and the Office-31 [11] dataset serves as one domain (Caltech). We implement the ResNet in FL

DomainNet contains six domains: clip art, quick drawing, real scene, painting, infographic and sketch. We implement the DenseNet in FL.

Digit recognition task: MNIST [38] and SVHN [24], MNIST is composed of numbers written by different people (integer 0-9). SVHN is a dataset that identifies Arabic numerals in the house number image. To make the format consistent, we align the size of MNIST with SVHN, and adjust MNIST from 28∗28∗1 to 32∗32∗3, repeat gray to RGB. We implement the AlexNet [41] as the base model in this experiment and set the learning rate as η_w=0.001.

In this paper, we compare the performance of the proposed FM^2 with State-of-the-Art (SOTA) and only report the MDMT scenario, where the baseline include FedProx [13], FedAvg [26] and SingleSet [23] (that is, using a single data set to train the model independently on each client)), SOTA (MOCHA [31] and FedDGA [33]) using customized methods on multi-task.

The experimental configurations of different schemes are summarized in Table VI, where $\star$ denotes a mixed dataset containing 50% MNIST and 50% SVHN. The experimental results for digits recognition are reported in Table VII, including both average accuracy and the worst-case accuracy for each node. For SDST, although centralized models achieve good performance on their own domain, cross-domain generalization drops dramatically (e.g., node 1 trained on MNIST only achieves 0.3073 when tested on SVHN). According to the results, SDMT consistently achieves slightly higher accuracy than MDST (0.9362 vs. 0.9305 on MNIST, and 0.9566 vs. 0.9524 on SVHN), as federated aggregation in SDMT better utilizes task-specific knowledge across clientsFM^2 further improves performance by simultaneously addressing task and domain heterogeneity, while MDMT benefits significantly from data accumulation and cross-domain collaboration, achieving the best overall accuracy.

The comparative study for various federated learning on the Office-Caltech-10 and DomainNet datasets is displayed in Table VIII. On most datasets and domains, our suggested FM^2 performs better than the current state-of-the-art techniques. FedProx functions similarly to FedAvg and augments it with additional terms. While both approaches exceed SingleSet on the Office-Caltech-10, in certain domains they might not match SingleSet's performance on the other two more difficult datasets. It is known that SingleSet and FedProx outperform FedAvg; nevertheless, only baseline findings are obtained because their primary purpose is not multi-domain FL design. On the other hand, FedDGA works better than the other

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TABLE VIII
TEST ACCURACY OF DIFFERENT METHODS

	Domains	SingleSet	FedAvg	FedProx	MOCHA	FedDGA	FM^2
Caltech-10	Amazon	0.5437	0.6142	0.5955	0.6232	**0.6713**	0.6702
	Webcam	0.8833	0.8254	0.8074	0.7896	0.8605	**0.9117**
	DSLR	0.8132	0.7703	0. 7662	0.7952	0.8654	**0.9422**
	Caltech	0.4031	0. 4408	0.4495	0. 4330	0.4542	**0.5074**
DomainNet	Clipart	0.4252	0.4867	0.5160	0.4921	0.4953	**0.5794**
	Infograph	0.2300	0.2565	0.2423	0.2456	**0.2830**	0.2803
	Painting	0.3502	0.3743	0.3721	0.3883	0.4075	**0.4359**
	Real	0.5262	0.4718	0.4675	0.4742	0.5415	**0.5560**
	Sketch	0.3334	0.3697	0.3895	0.3804	0.3962	**0.5081**
	Quickdraw	0.5556	0.4575	0.4678	0.4733	0. 6820	**0.6872**

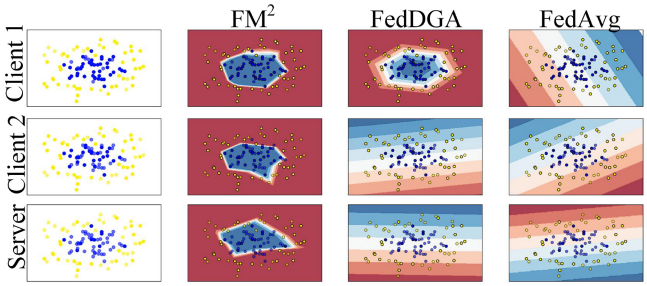

Fig. 5. Weight visualization of one convolutional layer in ResNet.

techniques and is intended for multi-domain FL. Furthermore, we discover that using a straightforward multi-task strategy (FedDGA) improves FedAvg's performance because it doesn't rely on batch statistics peculiar to a given domain.

Notably, our suggested FM^2 performs better on average accuracy across all datasets than both FedDGA and MOCHA.In certain domains, FM^2 surpasses FedDGA by more than 15%, while in other domains it falls behind by less than 1%. These findings show how useful FM^2 is when dealing with multiple tasks and domains.

C. Weight Visualization

As shown in Fig. 5, FedAvg and FedDGA exhibit notable discrepancies between client-local weight maps, with scattered activations suggesting domain-specific overfitting. In contrast, FM^2 produces highly similar weight structures across clients, as confirmed by an inter-client cosine similarity of 0.92, surpassing FedDGA (0.85) and FedAvg (0.71). These results quantitatively and visually confirm that FM^2 achieves better feature alignment and domain robustness.

D. Cyber Intrusion Detections

For anomaly detection in cyber security technology in an era of big data, anomaly detection or unexpected behavior can be detected by modelling the normal behavior of the system or network. KDD Cup′99 dataset is the most widely used benchmark dataset in the research of cyber intrusion anomaly detection. KDD Cup′99 dataset can be divided into four anomaly attack domain in more detail, namely Dos, R2L, U2R and Probe. To our best of knowledge, most of the current machine learning treat these different domain of attacks indiscriminately as an attack, then use a single algorithm for detection. In fact, however, a single attack classifier only has a high detection rates of attacks in one domain, and attacks against other domains are difficult to detected. Therefore, achieving multi-domain of anomaly detection is an effective way to solve this problem [15].

In order to simulate the real-world of Internet, we set up a training set of 95% normal and 5% abnormal in each domain, which is 20964 in Dos, 21699 in Probe, 1924 in U2R and 18604 in R2L. Similarly, in the test set, there are 14258 (including 5% abnormal) in the three domains of Dos, R2L and Probe, and 1400 in U2R (including 5% abnormal data). It should be pointed out that the KDD Cup′99 dataset has fewer data in U2R domain, so it has fewer in our training and test set. Multi-task multi-domain machine learning model we use is Autoencoder [32]. We also performed some preprocessing for dataset, for instance, text conversion to numeric, onehot and normalization. Moreover, since the amount of normal and abnormal data in each domain is a larg difference, the data on any node of our FM^2 learning cannot represent the characteristics of all data. Therefore, in this experiment, we also utilize GAN [22] to generate the same amount of abnormal as normal data.

The configuration scheme for this experiment is shown in Table IX, and $*$ define as 30% of Dos, 30% of R2L, 30% of Probe and 10% of U2R, $\diamond$ define as 50% of Dos and 50% of R2L, $\triangleright$ defne as 90% of Probe and 10% of U2R. As shown in Table X, SDST F1 is particularly inferior to the other three schemes. We believe that this is due to the fact that the amount of data onto a single node is small, so that machine learning cannot thoroughly learn the data specific of different tasks in different domains. SDMT and MDST have similar performance, as we explained in digits recognition experiment, we need to emphasize that the poor performance of R2L and U2R in this scheme is due to the similar distribution of R2L with normal data, therefore, the model is not easy to learn its intrusion characteristics, while U2R is because the amount of data is only dozens, it is also hard learning for the model. Surprisingly, although there is no data advantage in some domain, MDMT can greatly improve the performance of each domain, this is because it perfectly learns the specifics of different task in different domain through data accumulation, automatically compensates the performance of each domain. It also a perfect illustration of the ability of FL to handle multi-tasking multi-domain.

E. Robustness and Scalability

The outcomes of utilizing various small batchsize settings (batchsizes of 1 and 2, respectively) in the DomainNet

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TABLE IX
INTRUSION DETECTION EXPERIMENTAL CONFIGURATION

	node 1	node 2	node 3	node 4	node 5	node 6	node 7	federated
SDST	Dos	U2R	R2L	Probe	-	-	-	no
SDMT	Dos	U2R	R2L	Probe	-	-	-	yes
MDST	$*$	-	-	-	-	-	-	no
MDMT	Dos	U2R	R2L	Probe	$*$	$\diamond$	$\triangleright$	yes

TABLE X
INTRUSION DETECTION F1-SCORE

method	Dos		R2L		U2R		Probe	
	average	worst	average	worst	average	worst	average	worst
SDST	0.6139	0.4866	0.4172	0.2807	0.4507	0.3234	0.5652	0.4106
SDMT	0.9453	0.9311	0.8276	0.8106	0.8325	0.8233	0.9289	0.9092
MDST	0.9321	0.9291	0.8157	0.8004	0.8237	0.8185	0.9119	0.8979
MDMT	**0.9658**	0.9467	**0.9642**	0.9521	**0.9160**	0.8842	**0.9706**	0.9322

TABLE XI
PERFORMANCE USING SMALL BATCHES

	batchsize=1				batchsize=2			
Domains	SingleSet	FedAvg	FedProx	FM^2	SingleSet	FedAvg	FedProx	FM^2
Clipart	0.4032	0.4234	0.5006	0.4781	0.4131	0.4253	0.5123	0.4957
Infograph	0.2120	0.2235	0.2333	0.4531	0.2253	0.2414	0.2547	0.4563
Painting	0.3365	0.3413	0.3146	0.4429	0.3951	0.3532	0.3723	0.4698
Real	0.5032	0.4415	0.4778	0.4565	0.5332	0.4481	0.4816	0.5577
Sketch	0.3021	0.3397	0.3680	0.4431	0.3340	0.3507	0.3839	0.4815
Quickdraw	0.5052	0.4625	0.4138	0.4524	0.4532	0.4834	0. 4205	0.5079

dataset are displayed in Table XI. Table XI illustrates how our suggested FM^2 performs better than the current state-of-the-art techniques. While FedAvg, SingleSet, and FedProx yield comparable results when the batchsize is 1, FM^2 does not have these restrictions; instead, it does not necessitate the additional computation of the running mean and variance during the inference process. For cross-device FL, it is especially crucial that FM^2 outperforms in small batchesize, since certain edge devices could only be able to train in small batchesize when resources are limited.

We assessed the effect of randomly choosing a subset of clients to take part in every training session in order to compare the scalability of various FL. Since not every customer is able to attend every training session, this is a somewhat regular occurrence in cross-device FL. For our experiments, we randomly picked 5, 10, and 20 clients (i.e., K = 5, 10, 20 clients) to participate in each training round out of the 50 clients on the Kddcup'99 dataset. Our suggested FM^2 performs better than FedAvg, MOCHA, and FedDGA for all client count settings, as Table XII demonstrates.The selected adversarial training model, adaptive network topology, and distributed resilient optimization technique are responsible for this outperformance.

TABLE XII
RANDOMIZED CLIENT SELECTION

Client	Methods	Dos	R2L	U2R	Probe
5	SingleSet	0.5912	0.4234	0.4315	0.5736
	FedAvg	0.6453	0.6277	0.6355	0.6239
	FedProx	0.7321	0.7152	0.7211	0.8123
	MOCHA	0.7678	0.7686	0.7163	0.7705
	FedDGA	0.8321	0.8921	0.8157	0.8004
	FM^2	**0.9454**	**0.9261**	**0.9233**	**0.9490**
10	SingleSet	0.6233	0.5341	0.5166	0.6623
	FedAvg	0.6723	0.6748	0.6587	0.6934
	FedProx	0.7612	0.7577	0.7930	0.8302
	MOCHA	0.7814	0.8126	0.7903	0.7895
	FedDGA	0.8923	0.9214	**0.9537**	0.9042
	FM^2	**0.9522**	**0.9456**	0.9308	**0.9671**
20	SingleSet	0.6207	0.5423	0.5672	0.6521
	FedAvg	0.6802	0.6834	0.6457	0.7312
	FedProx	0.7654	0.7729	0.8030	0.8423
	MOCHA	0.8076	0.8366	0.8339	0.8532
	FedDGA	0.9012	0.9346	**0.9677**	0.9213
	FM^2	**0.9556**	**0.9514**	0.9511	**0.9663**

VI. CONCLUSION

In this paper, we propose the FM^2 Learning framework, a novel approach that combines LLMs with a federated multi-task and multi-domain optimization strategy, specifically designed to cope with task heterogeneity, domain migration, and edge resource constraints faced in consumer electronics and IoT environments. Through modular model design, two-stage optimization strategy, and robust optimization mechanisms, FM^2 achieves a good balance between personalization and generalization performance.

Experimental validation shows that FM^2 has significant performance improvement over existing methods on multiple standard datasets, especially in non-IID, multi-task, and multi-domain complex environments, where it exhibits excellent robustness and scalability. Meanwhile, FM^2 demonstrates the feasibility of embedding the capabilities of lightweight LLMs into federated learning systems, providing a new paradigm for building next-generation smart terminals.

REFERENCES

[1] A. Abboud, M.-E.-A. Brahmia, A. Abouaissa, A. Shahin, and R. Mazraani, "A hybrid aggregation approach for federated learning to improve energy consumption in smart buildings," in *Proc. Int. Wireless Commun. Mobile Comput. (IWCMC)*, 2023, pp. 854–859.

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[2] N. K. Agrawal et al., "TFL-IHOA: Three-layer federated learning-based intelligent hybrid optimization algorithm for Internet of Vehicle," *IEEE Trans. Consum. Electron.*, vol. 70, no. 3, pp. 5818–5828, Aug. 2024.

[3] Z. B. Akhtar, "Unveiling the evolution of generative AI (GAI): A comprehensive and investigative analysis toward LLM models (2021–2024) and beyond," *J. Electr. Syst. Inf. Technol.*, vol. 11, no. 1, p. 22, 2024.

[4] R. Al-Huthaifi, T. Li, W. Huang, J. Gu, and C. Li, "Federated learning in smart cities: Privacy and security survey," *Inf. Sci.*, vol. 632, pp. 833–857, 2023.

[5] A. Alzahrani and M. Z. Asghar, "Maintaining user security in consumer electronics-based online recommender systems using federated learning," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2657–2665, Feb. 2024.

[6] A. K. Bashir et al., "Federated learning for the healthcare metaverse: Concepts, applications, challenges, and future directions," *IEEE Internet Things J.*, vol. 10, no. 24, pp. 21873–21891, Dec. 2023.

[7] R. Berriel et al., "Budget-aware adapters for multi-domain learning," 2019, *arXiv:1905.06242*.

[8] K. Bonawitz et al., "Towards federated learning at scale: System design," 2019, *arXiv:1902.01046*.

[9] R. Caruana, "Multitask learning," *Mach. Learn.*, vol. 28, no. 1, pp. 41–75, 1997.

[10] A. Chang et al., "BIPEFT: Budget-guided iterative search for parameter efficient fine-tuning of large pretrained language models," in *Proc. Find. Assoc. Comput. Linguist.*, 2024, pp. 7429–7440.

[11] C. Chen et al., "Trustworthy federated learning: Privacy, security, and beyond," *Knowl. Inf. Syst.*, vol. 67, no. 3, pp. 2321–2356, 2025.

[12] R. S. Chen, B. Lucier, Y. Singer, and V. Syrgkanis, "Robust optimization for non-convex objectives," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 4705–4714.

[13] R. Chen, D. Shi, X. Qin, D. Liu, M. Pan, and S. Cui, "Service delay minimization for federated learning over mobile devices," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 4, pp. 990–1006, Apr. 2023.

[14] K. Cheng, T. Fan, Y. Jin, Y. Liu, T. Chen, and Q. Yang, "SecureBoost: A lossless federated learning framework," 2019, *arXiv:1901.08755*.

[15] M. Cho, T. Kim, M. Shim, D. Wee, and S. Lee, "Towards multi-domain learning for generalizable video anomaly detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 50256–50284.

[16] X.-Y. Deng and Y.-M. Fan, "Improved gradient descent optimizer with adaptive parameter setting and swarm intelligence," *J. Comput.*, vol. 35, no. 3, pp. 107–127, 2024.

[17] S. F. dos Santos, R. Berriel, T. Oliveira-Santos, N. Sebe, and J. Almeida, "Budget-aware pruning for multi-domain learning," in *Proc. Int. Conf. Image Anal. Process.*, 2023, pp. 477–489.

[18] M. Dredze, A. Kulesza, and K. Crammer, "Multi-domain learning by confidence-weighted parameter combination," *Mach. Learn.*, vol. 79, nos. 1–2, pp. 123–149, 2010.

[19] Meta Fundamental AI Research Diplomacy Team (FAIR) et al., "Human-level play in the game of diplomacy by combining language models with strategic reasoning," *Science*, vol. 378, no. 6624, pp. 1067–1074, 2022.

[20] H. Gao, B. Qiu, Y. Wang, S. Yu, Y. Xu, and X. Wang, "TBDB: Token bucket-based dynamic batching for resource scheduling supporting neural network inference in intelligent consumer electronics," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 1134–1144, Feb. 2024.

[21] Y. Goh, D. Jung, G. Hwang, and J.-M. Chung, "Consumer electronics product manufacturing time reduction and optimization using ai-based PCB and VLSI circuit designing," *IEEE Trans. Consum. Electron.*, vol. 69, no. 3, pp. 240–249, Aug. 2023.

[22] I. J. Goodfellow et al., "Generative adversarial nets," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2014, pp. 1–9.

[23] D. Javeed, M. S. Saeed, P. Kumar, A. Jolfaei, S. Islam, and A. N. Islam, "Federated learning-based personalized recommendation systems: An overview on security and privacy challenges," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2618–2627, Feb. 2024.

[24] A. Koloskova, N. Loizou, S. Boreiri, M. Jaggi, and S. Stich, "A unified theory of decentralized SGD with changing topology and local updates," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 5381–5393.

[25] W. Kuang et al., "FederatedScope-LLM: A comprehensive package for fine-tuning large language models in federated learning," in *Proc. 30th ACM SIGKDD Conf. Knowl. Discov. Data Min.*, 2024, pp. 5260–5271.

[26] N. Lin, Y. Wang, E. Zhang, K. Yu, L. Zhao, and M. Guizani, "Feedback delay-tolerant proactive caching scheme based on federated learning at the wireless edge," *IEEE Netw. Lett.*, vol. 5, no. 1, pp. 26–30, Mar. 2023.

[27] F. Liu, H. Wang, M. Jiang, and F. Zhang, "Balance-blended adversarial distribution and smooth-suppressed labels refinement network for partial transfer fault diagnosis," *Eng. Appl. Artif. Intell.*, vol. 135, Sep. 2024, Art. no. 108723.

[28] P. Liu, Q. He, Y. Chen, S. Jiang, B. Zhao, and X. Wang, "A lightweight authentication and privacy-preserving aggregation for blockchain-enabled federated learning in VANETs," *IEEE Trans. Consum. Electron.*, vol. 71, no. 1, pp. 1274–1287, Feb. 2025.

[29] N. Mahla, K. S. Jadhav, and G. Ramakrishnan, "Exploring gradient subspaces: Addressing and overcoming LoRA's limitations in federated fine-tuning of large language models," 2024, *arXiv:2410.23111*.

[30] F. Morandin, G. Amato, R. Gini, C. Metta, M. Parton, and G.-C. Pascutto, "SAI a sensible artificial intelligence that plays go," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, 2019, pp. 1–8.

[31] B. Petrovska, T. Atanasova-Pacemska, R. Corizzo, P. Mignone, P. Lameski, and E. Zdravevski, "Aerial scene classification through fine-tuning with adaptive learning rates and label smoothing," *Appl. Sci.*, vol. 10, no. 17, p. 5792, 2020.

[32] Q. Qian, S. Zhu, J. Tang, R. Jin, B. Sun, and H. Li, "Robust optimization over multiple domains," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 4739–4746.

[33] P. Rani et al., "Federated learning-based misbehavior detection for the 5G-enabled Internet of Vehicles," *IEEE Trans. Consum. Electron.*, vol. 70, no. 2, pp. 4656–4664, May 2024.

[34] X. Ruan, P. Jiang, Q. Zhou, J. Hu, and L. Shu, "Variable-fidelity probability of improvement method for efficient global optimization of expensive black-box problems," *Struct. Multidiscipl. Optim.*, vol. 62, no. 6, pp. 3021–3052, 2020.

[35] S. Ruder, "An overview of gradient descent optimization algorithms," 2016, *arXiv:1609.04747*.

[36] O. Sener and V. Koltun, "Multi-task learning as multi-objective optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 527–538.

[37] R. Senthil, B. Narayanan, and K. Velmurugan, "Develop the hybrid adadelta stochastic gradient classifier with optimized feature selection algorithm to predict the heart disease at earlier stage," *Meas., Sens.*, vol. 25, Feb. 2023, Art. no 100602.

[38] M. Shahraeini and R. Soltanifar, "A complex network-based approach for designing of wide area measurement systems in smart grids using Adam-eve like genetic algorithm," *Int. J. Eng.*, vol. 37, no. 2, pp. 298–311, 2024.

[39] M. Shin, J. Kim, B. Van Opheusden, and T. L. Griffiths, "Superhuman artificial intelligence can improve human decision-making by increasing novelty," *Proc. Nat. Acad. Sci.*, vol. 120, no. 12, 2023, Art. no. e2214840120.

[40] Z. Sun et al., "A survey on federated recommendation systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 6–20, Jan. 2025.

[41] J.-H. Syu, J. C.-W. Lin, G. Srivastava, and K. Yu, "A comprehensive survey on artificial intelligence empowered edge computing on consumer electronics," *IEEE Trans. Consum. Electron.*, vol. 69, no. 4, pp. 1023–1034, Nov. 2023.

[42] J. Wang et al., "Cooperative and competitive multi-agent systems: From optimization to games," *IEEE/CAA J. Automatica Sinica*, vol. 9, no. 5, pp. 763–783, May 2022.

[43] F. Wu, Z. Li, Y. Li, B. Ding, and J. Gao, "FedBIoT: LLM local fine-tuning in federated learning without full model," in *Proc. 30th ACM SIGKDD Conf. Knowl. Discov. Data Min.*, 2024, pp. 3345–3355.

[44] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, p. 12, 2019.

[45] Y. Yang and T. M. Hospedales, "A unified perspective on multi-domain and multi-task learning," 2014, *arXiv:1412.7489*.

[46] R. Ye et al., "FedLLM-bench: Realistic benchmarks for federated learning of large language models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 111106–111130.

[47] C. Zhang, G. Shan, and B.-H. Roh, "FMD-IoV: Security and robust enhancement for federated multi-domain learning–based IoV," *IEEE Trans. Intell. Transp. Syst.*, early access, Jan. 23, 2025, doi: 10.1109/TITS.2025.3527455.

[48] Y. Zhang and Q. Yang, "An overview of multi-task learning," *Nat. Sci. Rev.*, vol. 5, no. 1, pp. 30–43, 2018.

[49] H. H. Zhuo, W. Feng, Q. Xu, Q. Yang, and Y. Lin, "Federated reinforcement learning," 2019, *arXiv:1901.08277*.

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